Detecting Ground Deformation in the Built Environment Using Sparse Satellite InSAR Data With a Convolutional Neural Network

Nantheera Anantrasirichai, *Member, IEEE*, Juliet Biggs, Krisztina Kelevitz, Zahra Sadeghi, Tim Wright, James Thompson, Alin Marian Achim, *Senior Member, IEEE*, and David Bull, *Fellow, IEEE*

***Abstract*— The large volumes of Sentinel-1 data produced over Europe are being used to develop pan-national ground motion services. However, simple analysis techniques like thresholding cannot detect and classify complex deformation signals reliably making providing usable information to a broad range of nonexpert stakeholders a challenge. Here, we explore the applicability of deep learning approaches by adapting a pretrained convolutional neural network (CNN) to detect deformation in a national-scale velocity field. For our proof-of-concept, we focus on the U.K. where previously identified deformation is associated with coal-mining, ground water withdrawal, landslides, and tunneling. The sparsity of measurement points and the presence of spike noise make this a challenging application for deep learning networks, which involve calculations of the spatial convolution between images. Moreover, insufficient ground truth data exist to construct a balanced training data set, and the deformation signals are slower and more localized than in previous applications. We propose three enhancement methods to tackle these problems: 1) spatial interpolation with modified matrix completion; 2) a synthetic training data set based on the characteristics of the real U.K. velocity map; and 3) enhanced overwrapping techniques. Using velocity maps spanning 2015–2019, our framework detects several areas of coal mining subsidence, uplift due to dewatering, slate quarries, landslides, and tunnel engineering works. The results demonstrate the potential applicability of the proposed framework to the development of automated ground motion analysis systems.**

***Index Terms*— Convolutional neural network (CNN), earth observation, ground deformation, interferometric synthetic aperture radar (InSAR), machine learning.**

I. Introduction

FOR the last few decades, it has been possible to accurately measure ground deformation from space using interferometric synthetic aperture radar (InSAR) [1]. Recent advances in processing techniques and computing power [2], coupled with the launch of the Sentinel-1 satellites have laid the foundation for millimeter-scale monitoring of ground deformation across Europe in near real time. This has obvious potential for monitoring ground movement in urban and semirural environments. We use the U.K. as a test case, where the average annual cost to the insurance industry of ground motion is estimated to be over £250M [3], [4]. Incidents affecting critical infrastructure, such as mainline railways or dams, can be associated with multimillion pound costs, even for a single slope failure event. The sources of deformation are both natural and anthropogenic: subsidence and heave due to the legacy of the coal mining and quarrying industries [5], shrink and swell of shallow clays [6], natural sinkholes [7], landslides [8], coastal erosion [9], and engineering work, such as tunneling [10].

The Sentinel-1 satellites acquire data over a 250-km swath at a 4 m $\times$ 14 m spatial resolution every six days on both ascending and descending tracks, generating a large quantity of data. So far, efforts have largely focused on improving data processing methods and capacity [11], but the need for manual inspection and expert interpretation are also barriers to the timely dissemination of information. Various approaches to automatic detection have been tested, for example, Raspini *et al.* [12] use a threshold of 10 mm/year to identify anomalies in time-series data from Northern Italy. However, applying a threshold in the spatial domain is not reliable due to the effect of reference-point selection and the performance deteriorates heavily for noisy and low coherence signals. Albino *et al.* [13] used receiver operating characteristics to demonstrate that applying a cumulative sum control chart [14] to the time-series improves detection performance over simple thresholding. However, both these methods work on individual pixels and do not take into account the high spatial resolution information that is a major advantage for InSAR. Independent component analysis (ICA) has been used to separate deformation from noise based on the assumptions that the signals are statistically independent and non-Gaussian [15]–[17]. However, the main drawback for the use of ICA in automated systems is the uncertainty in the order of the separated components, known as the permutation problem [18].

In this article, we employ a convolutional neural network (CNN) to automatically detect ground deformation across the U.K. Deep learning has been employed in several acquisition types of remote sensing for nearly a decade, e.g., very

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Nantheera Anantrasirichai, Alin Marian Achim, and David Bull are with Visual Information Laboratory, University of Bristol, Bristol BS1 5DD, U.K. (e-mail: n.anantrasirichai@bristol.ac.uk; dave.bull@bristol.ac.uk).

Juliet Biggs is with the School of Earth Sciences, University of Bristol, Bristol BS8 1RJ, U.K. (e-mail: juliet.biggs@bristol.ac.uk).

Krisztina Kelevitz, Zahra Sadeghi, Tim Wright, and James Thompson are with the COMET, School of Earth and Environment, University of Leeds, Leeds LS2 9JT, U.K. (e-mail: k.kelevitz@leeds.ac.uk; z.sadeghi@leeds.ac.uk; t.j.wright@leeds.ac.uk).

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high resolution (VHR) satellite images [19] and hyperspectral images [20], [21]. Here, InSAR images are employed and the CNN models the spatial characteristics of the InSAR data and then recognizes the difference between deformation and atmospheric noise. We base our study on a transferable machine learning approach that has already been successfully used for detecting volcanic deformation in global InSAR data [22]–[24]. Adapting these approaches for detecting urban deformation is conceptually straightforward, but challenging to implement due to the unsuitable nature of available signals for CNN-based algorithms. The sources of deformation in the U.K. are much shallower and slower than in volcanic environments, meaning the deformation has a smaller magnitude and spatial extent. The spatially variable coherence and associated processing methods means that InSAR data for the U.K. is typically sparse and has different noise characteristics to volcanic environments.

In this article, we propose three novel contributions to address these problems: 1) spatial interpolation with a modified matrix completion (MC) method to tackle sparsity and simultaneously mitigate noise due to atmospheric effects and scatterer properties; 2) a new synthetic data set for training based on the characteristics of real U.K. velocity maps; and 3) enhanced overwrapping techniques with offset and gain to minimize the influence of reference point selection and to increase the likelihood of detecting slow deformation.

II. Materials and Methods

A. CNNs

CNNs are a class of deep feedforward artificial neural networks. They comprise a series of convolutional layers that are designed to take advantage of 2-D structures, such as an image. The weights of the filter in each convolutional layer are adjusted during the training process. The low-level features are extracted and connected to more semantic meaning at the deeper layers. In this article, we want to learn features from the velocity maps that can distinguish deformation from stable ground.

Previous studies have used CNNs to detect deformation in wrapped InSAR images of volcanic environments [22]–[24]. Wrapped interferograms are used because the high-frequency content of the fringes is easy to identify and provides strong features for the CNN. The work in [22] provided a proof of concept using a test data set of 30 249 interferograms, compared different pretrained networks and found AlexNet [25] to be the most effective and used data augmentation to train the network. The subsequent work [23] improved the detection performance by overcoming the lack of positive training data by using synthetic examples, representing deformation, turbulent, and stratified atmospheric contributions. Recently, we studied the feasibility of using the CNN to detect slow volcanic deformation by rewrapping cumulative time series [24]. We found that applying a gain of 2 to the interferograms to double the number of fringes can lower the detection threshold by 25%–30%, which can be as low as 1.3 cm/year.

In this article, we use a transfer-learning strategy augmented with fine-tuning the model trained in [23]. Then the CNN model is retrained with some negative samples of the real U.K. data along with synthetic positive and negative samples, based on the characteristics of the real U.K. data as described in Section III-B. In the prediction process, the velocity maps are wrapped and converted into a grayscale image (i.e., the pixel values are scaled to $[0, 255]$). Then they are divided into overlapping patches at the required input size for AlexNet (224×224 pixels). Each patch is then repeatedly shifted (by $28 = 224/8$ pixels in this article) to cover the entire image. The output of the prediction process is a probability P of there being deformation in each patch. The probabilities from overlapping patches are merged using a rotationally symmetric Gaussian low-pass filter with a size of 20 pixels and standard deviation of five pixels.

B. U.K. InSAR Data Set

Fundamentally, all InSAR methods use the phase difference between two radar images to estimate changes in path length between the satellite and the ground surface. However, there are two distinct classes of processing approaches for generating time series of data: small baseline and persistent scatterer (PS). The small baseline technique [26], [27] employs many small distributed scatterers and is commonly used for wide area monitoring, including tectonic and volcanic applications (e.g., http://comet.nerc.ac.uk/COMET-LiCS-portal/). It produces a series of 2-D images that can be straightforwardly employed by a CNN as shown in [24]. In contrast, permanent or PS methods [28], [29] focus on pixels dominated by a stable large reflector. Thus, PS methods are well-suited to areas that have strong reflectors, especially man-made objects like buildings and are usually preferred for urban areas [30]. However, the output data set is sparse and not suitable for input into CNNs, where correlations between adjacent pixels are learned and used as local features for classification.

The InSAR data set used in this article was provided by SatSense Ltd., who employed a novel pixel selection method, RapidSAR [2]. This technique works by identifying siblings of the selected pixel, i.e., evaluating nearby pixels with similar phase and amplitude to the selected pixel. This is then used to estimate the coherence of the selected pixel. This avoids the common issue with both PS and small baseline methods whereby coherent points may be rejected or incoherent points included, due to the effect of surrounding pixels. The associated information loss and lower SNR is therefore avoided. However, this still corresponds to sparse representation, which is not directly suitable for CNNs.

For the U.K.-wide study, we use the medium resolution SatSense product (10 m/pixel) for the period of 2015–2019 which consists of $66\,801 \times 121.501$ pixels. Although time series are available for each point, for this initial proof of concept, we simply use the average velocity for each pixel. In total, there are $\sim$64 million velocity measurements on the ascending pass and $\sim$29 million on the descending pass. The distribution of measurement locations is uneven with a significantly higher density in urban areas. We also identify three case study

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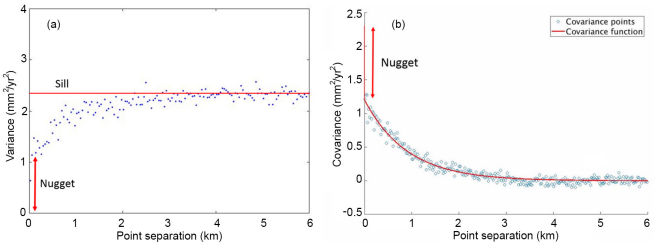


Fig. 1. Spatial characteristics of data in Easton, U.K. (a) 6-km length variogram, showing the nugget near 0, then levels off to a sill of 2.3 mm^2/year^2 at point separations of above around 2 km. (b) Covariance function, showing the exponential fitted to the data, the nugget at zero point separation, and the sill as the function levels off at point separations greater than 2 km.

areas from the high resolution SatSense product (5 m/pixel). The coal mining area of Normanton and Castleford shows subsidence of more than 2 mm/year [Fig. 5(a)] and South Derbyshire shows uplift of more than 6 mm/year [Fig. 5(d)]. A linear pattern of subsidence is seen from Battersea Power Station to Kennington in London [Fig. 5(g)]. This is the Northern line extension, where two 3.2-km tunnels have been created between 2017 and 2020. The difference between the two resolutions is illustrated in Fig. S1 in the Supplementary Material.

To analyze the spatial characteristics of the SatSense data sets in the U.K., we performed a spatial analysis using covariogram [31]. First, a spatial variogram $\gamma(d)$ for point velocity values in space is computed, where d is the distance between the pixels. We found that the variance of point velocity increases sharply (the nugget μ_{nugget}) when the distance between the points is close to zero, then exponentially increases and exhibits a sill μ_{fill}, the background variance value, at long length scales. Consequently, a theoretical variogram is related to covariance $C(d)$ on the basis of $\gamma(d) = \mu_{\text{sill}} - C(d)$. That is, the covariance $C(d)$ decreases exponentially when the distance between points is more than zero, expressed as

$$C(d) = \begin{cases} ae^{-bd}, & \text{if } d > 0 \\ \mu_{\text{sill}}, & \text{if } d = 0 \end{cases} \tag{1}$$

where a and b are constants, $a = \mu_{\text{sill}} - \mu_{\text{nugget}}$, and d is the separation distance in km. From the available U.K. data set, we found $a = 0.7 - 1.8$ mm^2/year^2, $b = 0.8 - 1.6$, and $\mu_{\text{sill}} = 1.5 - 2.9$ mm^2/year^2. This appears as spike noise in the InSAR image and disturbs the gradient calculations performed by the CNN. Thus, the spike noise needs to be accounted for when addressing the issue of data sparsity. The plots of the variogram and covariance are shown in Fig. 1.

III. Theoretical Contributions

A. Spatial Interpolation

CNNs rely on spatial or sequential attributes of dense data to learn effectively. Adjacent pixels share information that is important and the inherent structure to pixels in image data gives meaning to the overall image. If the data are highly sparse, then the network learns "zeros," the gradient of the loss function is zero and the performance does not improve with iteration. Therefore, it is necessary to interpolate the data during preprocessing to resemble a dense image. Here, we propose and test a novel interpolation method specifically for sparse InSAR data. We illustrate the process using the case study of Normanton and Castleford as shown in Fig. 2(a)–(c) and test the ability of the CNN to identify signals for different types of interpolation in Section IV-A.

The simplest way to mathematically describe sparse images is by $y = Mx + n$, where y is the sparse observation of an ideal dense signal x, M is the subsampling matrix, which can be seen as a mask of existing or nonexisting values, n is noise. Here, y is the raw velocity measurements shown in Fig. 2(a). This poses an inverse problem for finding x. We employ an MC method which has been used for compressive sensing [32], where the sparsity of a signal can be exploited to recover it from far fewer samples than required by the Nyquist Shannon sampling theorem [33]. This can be solved with an optimization process as

$$\hat{x} = \arg\min_x \left\{ \frac{1}{2}||y - Mx||_2^2 + \alpha||x||_* \right\} \tag{2}$$

where $||x||_*$ is the nuclear norm of a matrix (a convex hull of the rank function of x) and α is a regularization parameter. This can be done through a nonconvex MC via iterated soft thresholding [34]. The nuclear norm is computed using singular values of matrix x and the process tries to achieve

$$\min_x ||S_x||_p \text{ s. t. } ||y - Mx||_2 < \varepsilon \tag{3}$$

where $U_x, S_x, V_x = \text{SVD}(x)$, SVD is singular value decomposition giving the outputs such that $x = U_x S_x V_x'$ and for a nonconvex function, $0 < p < 1$. The pseudocode to describe this optimization process is given in Algorithm 1.

First, we generate an initial x_0 by first suppressing some high noise and then applying Delaunay triangulation (DT) [Fig. 2(b)]. To suppress the high-amplitude noise, we simply apply a 2-D median filter $\text{Med}_{3\times3}(\bullet)$ that omits NaN values in the median calculation. We record the noise map $N = y - \text{Med}_{3\times3}(y)$, which will be used later for generating synthetic data with similar characteristics (Section III-B).

In the interpolation process, we add a Gaussian filter $G(x, \sigma)$ with standard deviation σ of five pixels, to remove the remaining spike noise in each iteration loop. The proposed technique achieves the estimation of missing pixels and noise reduction simultaneously. The interpolated result is shown in Fig. 2(c). The wrapped version in Fig. 2(d) shows a clear pattern of fringes in contrast to the wrapped version of the raw data, which shows a noisy and unclear pattern and would be challenging to identify using any automated detection method [see the Supplementary Material (Fig. S2)]. Fig. 4 shows that the proposed MC method produces more realistic results than conventional DT alone.

B. Synthetic Examples

We create synthetic data sets (X) for training the CNN using two components, namely deformation D, and turbulent atmosphere T, using the simple linear function $X = D + T$.

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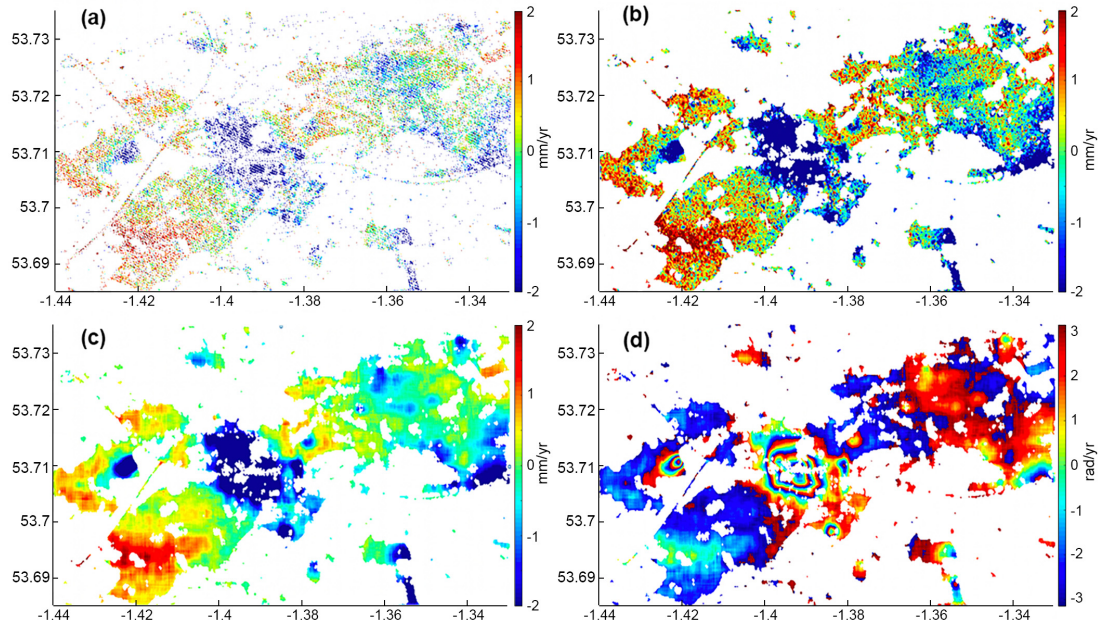


Fig. 2. Velocity map at Normanton and Castleford showing (a) raw sparse data, and its interpolated results from (b) DT and (c) MC techniques. The wrapped velocity map of (c) with the wrap gain μ=8 is shown in (d).

Algorithm 1 Pseudocode of Optimization Algorithm

Input: y, x_0, f_0, p, α_0, α, λ, τ, K

y : sparse observation
x_0 : interpolation using DT and noise suppression
M : sub-sampling matrix
α : regularization parameter, $\alpha_0 = 0.9\max(|Mx|)$
f: loss, initialed with $f_0 = ||y - Mx_0||_2 + \alpha_0||x_0||$
p : non-convex norm, $p = 0.8$
λ : 1.1·eigenvalue of $(M^{-1}M)$
τ : tolerance, $\tau = 10^{-4}$
C: cost function
K : maximum iterations, $K = 200$

Output: $\hat{x} = x$

while $\alpha > \tau\alpha_0$ **do**
  for $k = 1$ to K **do**
    $x \leftarrow x + \frac{1}{\lambda}M^{-1}(y - Mx)$
    $U, S, V \leftarrow \mathrm{SVD}(x)$
    $S \leftarrow \mathrm{diag}\{S\}$
    $S \leftarrow \mathrm{sign}(S)\max(0, |S| - \frac{1}{2\lambda}\alpha|S|^{p-1})$
    $x \leftarrow U(\mathrm{diag}\{S\})V'$
    $x \leftarrow G(x, \sigma)$
    $f_k \leftarrow ||y - Mx||_2 + \alpha||x||$
    $C \leftarrow ||f_k - f_{k-1}||/||f_k + f_{k-1}||$
    if $C < \tau$ **then**
      break
    end if
  end for
  $\alpha \leftarrow 0.9\,\alpha$
end while

Following our previous work [22], [23], each class has 10 000 samples which should be large enough for training [35]. Fig. 4 demonstrates the process of synthetic example generation for one example. In this article, we concentrate on deformation caused by coal mining and tunneling as they are common in the U.K. Therefore, we employ two models as follows.

1) A set of synthetic examples of coal mining subsidence, D_{point} is generated using a point pressure source model [36], which reproduces the surface deformation associated with inflation and deflation of a subsurface point source. To represent the shallow sources associated with coal mining, we use depths of 3–80 m and volume changes of $10^{0.3} - 10^3$ m^3.
2) A set of synthetic examples of tunneling subsidence, D_{line} is generated following [37]. The tunneling-induced subsidence profile is modeled with sagging and hogging zones as demonstrated in Fig. 3(a), where the length and depth parameters of sagging and hogging zones are l_{sag}, l_{hog}, d_{sag}, and d_{hog}, respectively.

We use both l_{sag} and l_{hog} in a range of 30–80 m, d_{sag} of 1–10 mm, and d_{hog} of 1–5 mm. D_{line} is generated by varying these parameters along the curve and straight lines, replicating the track of the underground tunnel. The 3-D displacement vector is then projected to line of sight (LOS) using Sentinel-1 U.K. incidence and heading angles for ascending and descending passes. For both cases, the range of parameters is chosen so that the LOS velocity is in the range 0–15 mm/year. Note that, in this paper, we trained the models of D_{point} and D_{line} separately, but they could be merged to train a 3-class model (two types of deformation and non-deformation) in the future.

The satellite measurements of displacement are affected by atmospheric delays caused primarily by water vapor in the troposphere, T. The delays are spatially correlated and

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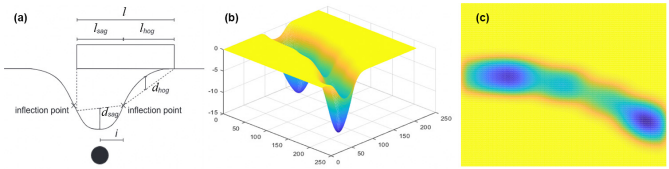


Fig. 3. Synthetic tunneling subsidence generated following the model introduced in [37], where the cross section profile is shown in (a). Our three-dimensional (3D) synthetic deformation and its projection to create two-dimensional (2D) unwrapped velocity map are shown in (b) and (c), respectively.

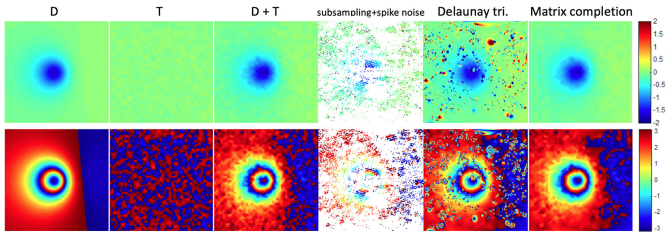


Fig. 4. Synthetic example showing (top row) unwrapped and (bottom row) wrapped samples.

their covariance is described in Section II-B. For simplicity, the statistical properties of the atmosphere are assumed to be radially symmetric and have a homogeneous structure [38]. We use Monte Carlo samples of these distributions to generate synthetic variance–covariance matrices and use a Cholesky decomposition to produce synthetic images with the corresponding statistical properties [23]. For previous applications to volcanic environments, we have also considered a stratified atmospheric component related to the high relief of volcanic edifices. This effect is small in the U.K. and is neglected here.

We then subsample the combined image $(D + T)$ using randomly chosen distributions of points from the SatSense data and add spike noise as described in Section II-B. This creates a noise model closer to the real data than simply using additive white Gaussian noise (AWGN). Finally, the sparse signals are interpolated as described in Section III-A.

C. Overwrapping and Phase Shifting

We wrap the velocity map to provide strong features for machine learning [22], i.e., edges where the phase jumps between $-\pi$ and π (the features activated by the first two convolutional layers are shown in the supplementary material (Fig. S3)). To deal with different deformation rates, we combine a range of wrapping intervals following the method of [24], which was originally designed to detect slow, sustained volcanic deformation in time-series data, but can be adapted for detecting slow, localized motion in the U.K. velocity measurements. Theoretically, the number of fringes can be increased without altering the signal to noise ratio by reducing the wrap interval (μ). In this paper, following Sentinel-1 LOS where one fringe represents 28 mm of displacement, we employ wrap intervals of 14 mm/year, 7 mm/year, 3.5 mm/year, and 1.75 mm/year in the prediction process.

One problem with wrapping the velocity map is that different reference points cause the wrap discontinuities to occur in physically arbitrary locations. For some choices of reference points, the number of fringes will increase, but for others it will decrease or for very small signals, fail to produce any discontinuities at all. To ensure that fringes exist on the test image, a constant offset τ is added to the velocity map ψ producing ψ'_τ, i.e. $\psi'_\tau \equiv \psi + \tau \mod \mu$. We run 4 offsets, and select the maximum probability from the CNN for each wrap interval μ, i.e. $P_\mu = \max\{P_{\mu,\tau}\}$, $\tau \in \{0, 3.5, 7, 10.5\}$ mm/yr, and $\mu \in \{14, 7, 3.5, 1.75\}$ mm/year. The final result is the average of the four probabilities, i.e., $P_{\text{final}} = (1/4) \sum_{\forall \mu} P_\mu$.

D. Combining Different LOS Geometries

One limitation of InSAR technology is that the ground motions are measured in a 1-D LOS geometry, whilst the actual surface motions can occur in three dimensions. This means the deformation detected in one LOS direction might not be able to be detected in another LOS direction. However, an advantage is that noise causing a false positive result that appears in one acquisition might not affect the acquisition in another LOS. Therefore, in this study, if the areas have both ascending and descending passes available, the two velocity maps are processed independently and the final probability results are obtained from the average. If there are four looks (two ascending and two descending passes), the final probability map will be the maximum of four averages between a pair of ascending and descending signals.

IV. Results and Discussion

A. Spatial Interpolation

We first investigate the performance of the proposed spatial interpolation technique using synthetic data sets. Three approaches are tested: 1) sparse examples without interpolation; 2) interpolated examples with DT; and 3) interpolated examples using the proposed MC approach (see Fig. 4 last three columns). The CNNs are trained with two classes: $D+T$ (positive) and T (negative). Each class contains 10 000 synthetic samples. When training the CNN with sparse examples, the results of convolution processes are computed from the pixels that have values only. The classification results are shown in Table I. It is obvious that without spatial interpolation, the CNN cannot distinguish between deformation and nondeformation (the accuracy is around 50%). The CNN performs significantly better with dense data sets with an improvement of accuracy by 64.0% with the initial DT and 81.5% with the proposed MC. The DT produces ten times more false positives than the MC due to spike noise (the nugget—see Section II-B).

B. Application to Case Study Sites

Initially, we test our machine learning algorithms on well-known case study examples of coalfield subsidence D_{point} and tunneling D_{line} (as described in Section III-B) using the high-resolution InSAR product (5 m/pixel). The models are trained separately, using the synthetic examples. The detection

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TABLE I
CLASSIFICATION PERFORMANCES (%) WHEN TRAINING WITH SPARSE AND INTERPOLATED EXAMPLES

Dataset	Accuracy	Precision	Recall	False positive rate
Sparse	54.32	63.91	53.62	55.27
Interp. DT	89.06	99.10	82.52	20.98
Interp. MC	98.58	99.27	97.93	2.09

results are shown in Fig. 5, where the first, the second and the third columns show: 1) raw InSAR data; 2) wrapped and interpolated velocity maps used as inputs of the CNNs; and 3) the probability values overlaid on the velocity maps, respectively. The first and the second rows are the results from the coalfields at Normanton and Castleford, and South Derbyshire, detected with the model D_{point}. The velocity map at South Derbyshire has fewer data points causing more difficulties for the interpolation step than that at Normanton and Castleford, but the detection algorithm still works well in both cases. This two study cases demonstrate that the CNNs can detect the fringe patterns that contain some missing areas. The last row of Fig. 5 shows the detected tunneling subsidence in London using the model D_{line}. Interestingly, the model detects the line of the tunnel but does not pick out the point-source deformation (on the right of the image). These case study results are promising and warrant further testing to check the generalization of the model and the applicability to a larger scale map.

C. Whole U.K. Velocity Map

As described in Section II-B, there are ~64 million points of sparse U.K. data. This is equivalent to a 2-D image with a resolution of 98 504 × 68 504 pixels, which is more than 3250 full HD TVs combined. To automatically process this large velocity map, we divide it into several 2500 × 2500 maps, defined by the limitation of memory required to process the spatial interpolation. After spatial interpolation, each velocity map is further divided into overlapping patches following the detection process described in Section II-A. The detected deforming locations using model D_{point} and D_{line} are plotted in Figs. 6 and 7, showing three levels of probability P, which are >0.5, >0.75 and >0.9. In the Supplementary Material (Figs. S4 and S5) we show areas with detection probabilities >0.5 in more detail.

Fig. 6 shows the results of the D_{point} model. The method detects numerous deforming areas in well-known coal-mining regions from the Midlands up toward Leeds (area A in Fig. 6), in South Wales (area B [39]), Normanton and Castleford (area C), North Staffordshire in Stoke-on-Trent (area D [40]), Northwest Leicester (area E [41]), Northumberland and Durham (area F [42]). Several areas are detected in London, where recent engineering work has taken place. For example, the detected uplift at Canning Town, London, could be affected by groundwater rebound after completion of dewatering works for the underground construction (area G [43]). In the northwest of Wales, the method detects subsidence from some former slate quarries (area I), including the Dinorwic Quarry near Llanberis [Fig. 6(c)], the Penrhyn Quarry near Bethesda, and the Ffestiniog Slate Quarry in Blaenau Ffestiniog, where the slate was mined rather than quarried. The method also detects subsidence of clay works in Kingsteignton [area J, Fig. 6(d)]. Uplift was detected at Golborne, Leigh and Manchester (area K) with a similar spatial extent to the subsidence reported between 1992 and 2000 [44]. Although we are dominantly considering vertical deformation, horizontal motion associated with landslides and coastal processes will also cause displacements in the line of sight (see Fig. S6 in the Supplementary Material). For example, landslides with significant horizontal motion were detected south of Kirkby Stephen (area H [45]).

We analyze our detection performance by plotting a histogram of the deformation rate against probability output of the detection method, shown in Fig. 8(a). Several areas have both high velocity and high probability as expected, confirming that both methods identify these regions. For example, areas D2 and Q2 in the Stoke-on-Trent [Fig. 8(b)] [40] and Lake District [Fig. 8(c)] [46] have velocities of -5.8 and -9.5 mm/year, respectively, and probabilities >0.95. However, the CNN assigns high probabilities to some areas with low-medium velocity because of their spatial pattern, for example D1 and Q1 [Fig. 8(b) and (c)] have velocities of 3.2 and 3.0 mm/year, respectively. These would have been missed with a simple thresholding technique if the threshold is set higher than 3.2 or 3.0 mm/year. The CNN also assigns low probabilities to some areas with high velocities because they do not have the appropriate spatial pattern. For example, Bristol city center has several pixels with velocities >5 mm/year which are caused by spiky noise that is assigned a low probability (<0.05). From this histogram, we set three thresholds at 3, 4.5, and 5 mm/year to classify slow, medium and fast deformation and apply a thresholding technique to the raw data. We apply a dilation morphological operation with a size of 18 pixels to connect sparse pixels (resulting 32 km^2 with minimum four velocity points) and also employ a 3×3 median filter to remove some spiky noise. Comparing with the proposed method, the thresholding technique generates significantly more positive points for expert to further investigate (258 versus 4064 positives). The detection results of the thresholding approach are shown in the supplementary material (Fig. S7 in the Supplementary Material).

Fig. 7 shows the results of the D_{line} model. We did not include examples of uplift in either positive or negative training data sets for D_{line}, but nonetheless, we detect several uplifting features because the fringes in the wrapped velocity map have characteristics closer to the positive samples than the negative ones. Since uplift and subsidence can be simply distinguished by comparing the velocity with that of neighboring areas, this information can be added in postprocessing. The only detection of tunneling subsidence in London was at the case study site shown in Fig. 5(g)–(i), but there were several detections elsewhere in the U.K., particularly in the Midlands. Several of these are elongated areas of subsidence more in keeping with mining (for example following coal seams) than infrastructure tunnels (area A in Fig. 7). In several cases, linear features are associated with linear surface structures, such as

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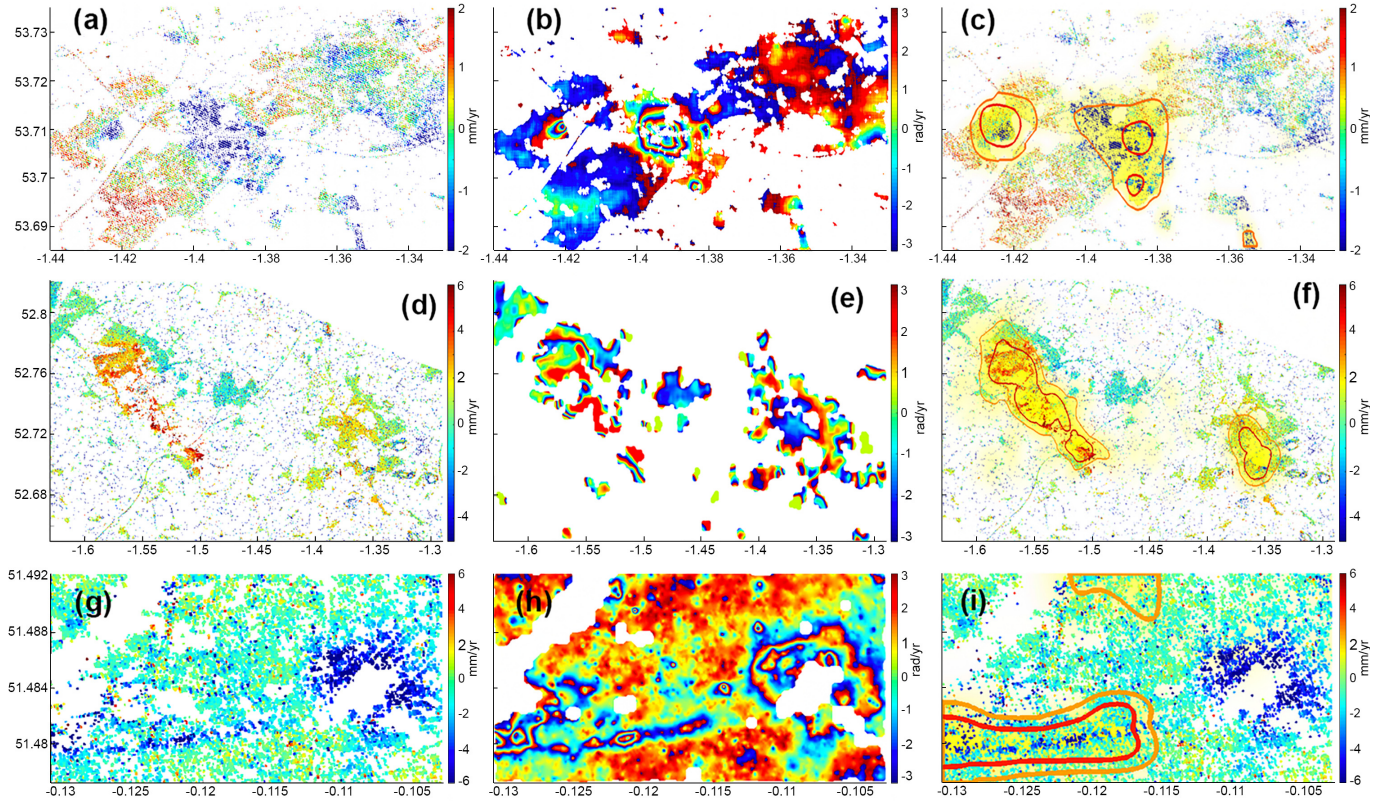


Fig. 5. Detection results in (a)–(c) Normanton and Castleford, (d) and (e) South Derbyshire, and (g)–(i) London—northern line extension. (a), (d), and (g) Raw data. (b), (e), and (h) Wrapped and interpolated velocity maps. (c), (f), and (i) Probability maps overlaid on the raw data. The brighter yellow means higher probability. Areas inside orange and red contours are where $P > 0.5$ and $P > 0.75$, respectively.

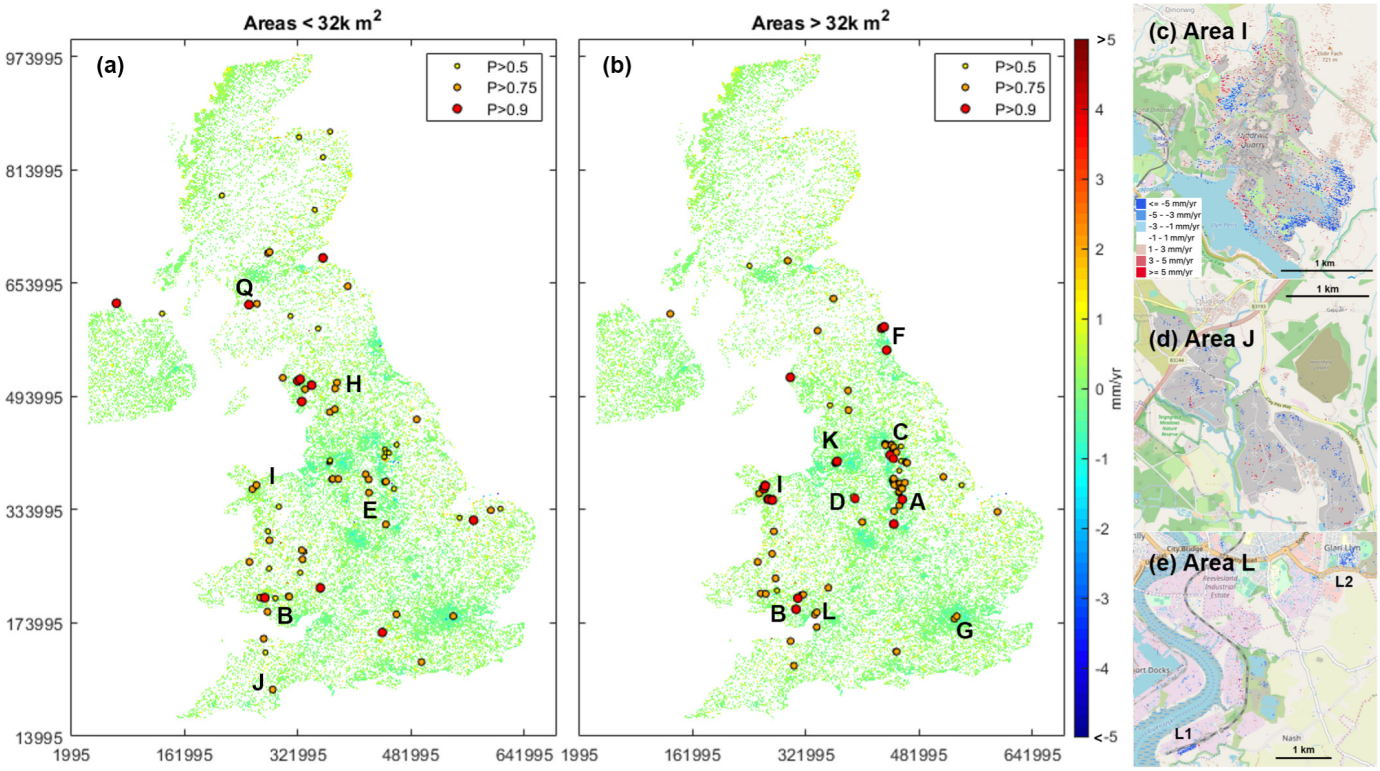


Fig. 6. Detection results in the 10-m velocity map showing the center locations where the CNN using D_{point} identifies with high probability of being deformation. For visualization, the small and large areas are plotted separately. (a) Area size less than 32 km^2. (b) Area size larger than 32 km^2. Right column shows ground subsidence due to anthropogenic sources at (c) Dinorwic quarry in North Wales (Area I), (d) clay works in Kingsteignton (Area J), and (e) coal yard of Uskmouth power station (L1) and residential areas around Brinell Square (L2) in Newport (Area L).

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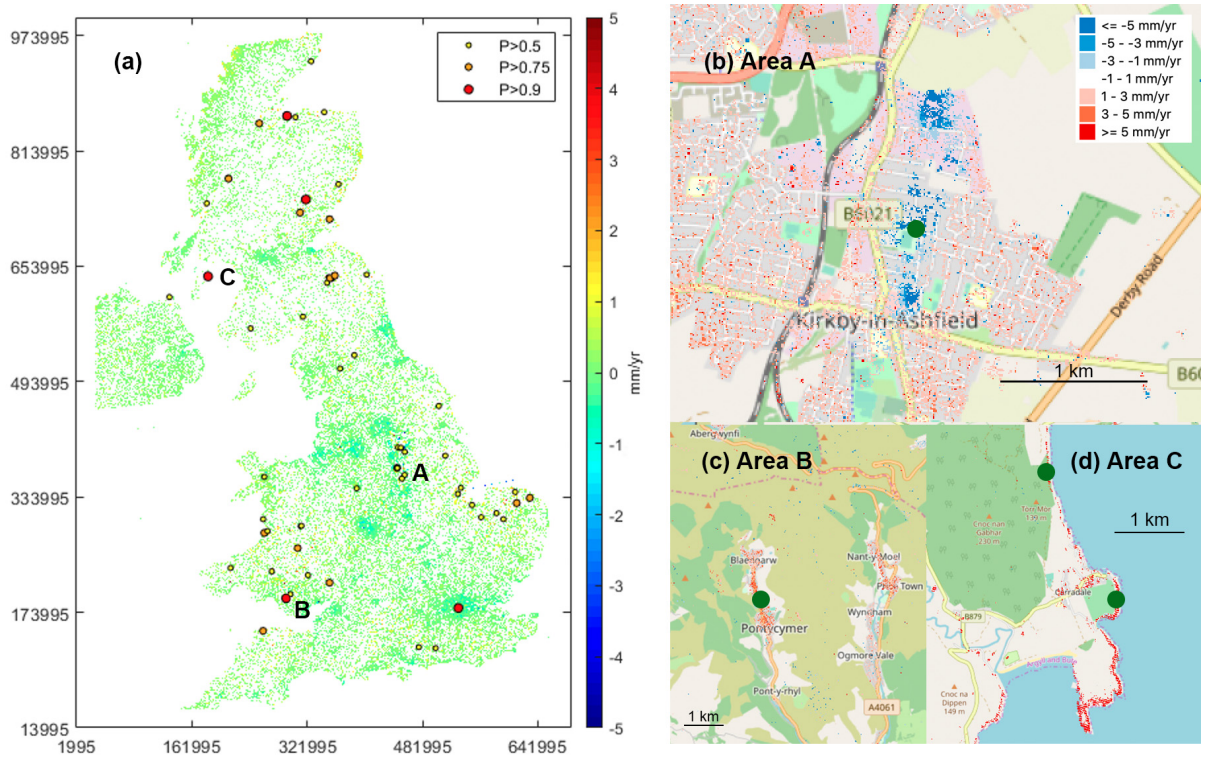


Fig. 7. Detection results in the 10-m velocity map (a) showing the center locations where the CNN using D_{line} identifies with high probability of being deformation. Examples of linear deformation in (b) Kirkby-in-Ashfield (Area A), (c) Pontycymer, Wales (Area B), and (d) uplift coastline in Carradale, Scotland (Area C). The green dots in (b)–(d) are the center of detected areas.

roads, probably due to the higher density of measurement points on the man-made structures than in the surrounding fields. The deformation signal itself is unlikely to be linear, but this enables us to identify deformation sources that might be missed by the d_{point} model due to the uneven sampling of data [e.g., Fig. 7(c)]. In several places, rocky foreshores are picked out, such as the coastline in Carradale [Fig. 7(d)], which appears to be uplifting relative to the nearest inland point. We attribute this to processing artifacts within the InSAR data.

D. Discussion

Monitoring ground deformation is crucial in urban and semi-urban areas. The U.K. has a long history of coal mining and associated water pumping causes surface deformation which can extend to city-sized areas. Slope instability can lead to localized damage both in hilly areas and coastal regions. Ground motion can have negative impacts on infrastructure, particularly long linear assets such as drainage networks and pipelines. An example of the need of ground movement detection is for the proposed HS2 route for high speed rail[1] from Birmingham to Leeds, which would pass through the large coalfield areas in Nottingham and Sheffield. Figs. 6 and 7 show clear ground deformation in these areas and although the velocity rate is only millimeters per year, it still needs to be factored into construction plans.

Previous work on deep learning has demonstrated its potential for automatically searching through large volumes of wrapped InSAR images to detect both slow and rapid ground deformation that may be related to volcanic activity [22]. In this article, we extend our work to the U.K. velocity map, where the measurement points are sparse and unevenly distributed. Moreover, the spatial noise characteristics of this data set are different from the distributed scatterer InSAR used in the volcano case. We analyzed this type of InSAR data and propose several new adaptations to allow the transfer learning approach to perform well under these circumstances.

This article is a proof-of-concept that demonstrates the potential applicability of the deep learning framework to the development of automated ground motion analysis for anthropogenic sources of deformation in urban and semiurban environments. We test the deep learning framework on the U.K. data set and produce a probability map of surface movement. As the data set is very large (see Section IV), it would not be feasible to manually inspect the entire area at high resolution. Using a probability threshold of 0.5, the method produces some false positives and false negatives. However, the probability values and the sizes of the detected areas can be employed to prioritize further analysis.

This approach is not restricted to the U.K. data set and could be used for any national or regional velocity map, including the European Ground Motion Service currently proposed by Copernicus [47]. The main limitation of the current framework is that it cannot detect very localized deformation, like sinkholes, because their spatial characteristics are too similar to noise. These areas, however, show clear changes in the time domain. Future developments can incorporate

[1]https://www.hs2.org.uk

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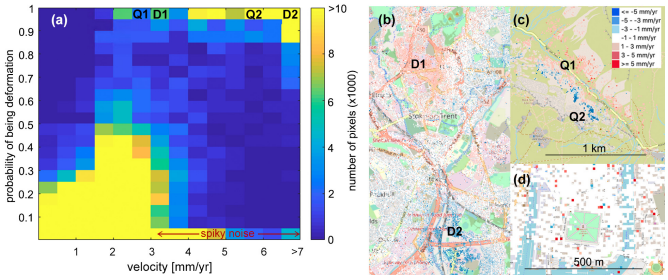


Fig. 8. Detection analysis of the D_{point} model. (a) Histogram of velocity against probability of being deformation. (b) North Staffordshire in stoke-on-trent (area D in Fig. 6). (c) Honister pass in Lake district (area Q in Fig. 6). (d) City of Bristol showing spiky noise in nonmoving area.

both time-series analysis and spatio-temporal (3-D) analysis of InSAR data. Moreover, if both ascending and descending passes are available for the same period of time, 4-D signals can be used. In this article, we train the model using only one pass (2-D), and the results of both passes are averaged (Section III-D). If both passes are concatenated and trained together, we expect that the deformation signals would be shown in both passes, so the number of false positives arising from using only one pass will be diminished.

V. Conclusion

This article demonstrates the feasibility of using a transferable CNN approach to detect ground deformation in urban and semiurban areas in the U.K. We analyze characteristics of the data and propose several adaptations to previously developed deep learning methods. MC is used to overcome the sparse and uneven measurement distribution and simultaneously reduce spike noise. Synthetic examples based on point sources and tunnels are used for training due to lack of real signals of deformation. Finally, overwrapping and phase shifting techniques are employed to enhance features and hence reduce the detection threshold. The methods are tested using the velocity map generated by SatSense Ltd. dated between 2015 and 2019 and successfully detect several types of deformation occurring around the U.K.

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Nantheera Anantrasirichai (Member, IEEE) received the B.E. degree in electrical engineering from Chulalongkorn University, Bangkok, Thailand, in 2000, and the Ph.D. degree in electrical and electronic engineering from the University of Bristol, Bristol, U.K., in 2007.

She is a Research Fellow with Visual Information Laboratory, University of Bristol. She has been involved in many projects across multiple disciplines, including telecommunications, biology, optometry, clinical sciences, biochemistry, mechanical engineering, psychology, and geoscience. She has published more than 50 articles. Her research interests include image and video coding, image analysis and enhancement, medical imaging, texture-based image analysis, machine learning, and remote sensing.



Juliet Biggs received the B.A. and M.Sci. degrees in natural sciences from the University of Cambridge, Cambridge, U.K., in 2003, and the Ph.D. degree from the University of Oxford, Oxford, U.K., in 2007, for her work using satellite radar to study the earthquake cycle in Alaska.

In 2010, she moved to the University of Bristol, Bristol, U.K., to take up a permanent lecturership and became a Full Professor in 2019. Over the last decade, she has studied Africa's Great Rift Valley, and the role played by volcanoes, magma intrusions, and faults during continental rifting. On a global scale, she was the first to quantify the link between deformation and eruption, the physical mechanisms responsible for coupled eruptions, and the effects of volatile content and phase changes on deforming systems. She has demonstrated that many volcanoes previously believed to be dormant are actually restless and provided the first satellite observations at several dangerous volcanoes in Latin America. Recently, she has worked closely with the Bristol Visual Information Laboratory to pioneer the use of deep learning for analyzing large InSAR data sets with applications to natural and anthropogenic sources of deformation.



Krisztina Kelevitz received the B.Sc. degree in Earth sciences (geophysics) from Eötvös Loránd University, Budapest, Hungary, in 2010, the M.Sc. degree in exploration geophysics from the University of Leeds, Leeds, U.K., in 2011, and the Ph.D. degree in geophysics (seismology and geodesy) from ETH Zürich, Zürich, Switzerland, in 2017.

She was a Post-Doctoral Research Fellow at CU Boulder, Boulder, CO, USA, from 2018 to 2019, and has been at the University of Leeds, Leeds, U.K., since 2019, working on the applications of InSAR remote sensing data in natural hazard monitoring. She is also working on enhancing InSAR data with corner reflectors for assessing structural stability of large infrastructure units (e.g., tunnels, bridges, and embankments).



Zahra Sadeghi received the M.S. and Ph.D. degrees in geomatics engineering-remote sensing from the K. N. Toosi University of Technology, Tehran, Iran, in 2011 and 2017, respectively, with a focus on developing performance of persistent scatterer InSAR (PSI) techniques for crustal deformation mapping.

From 2015 to 2017, she was a Visiting Ph.D. Student with the University College London, London, U.K., where she was involved in the combination of PSI and polarimetric coherence optimization methods. After completion of her Ph.D., she was employed by SatSense Ltd., Leeds, U.K., as an InSAR Developer. Since 2019, she has been with COMET, University of Leeds, Leeds, U.K., as a Post-Doctoral Research Fellow to work on a new algorithm for intercomparison of InSAR products and also high-resolution InSAR algorithms for infrastructure applications. Her research interests include developing PS InSAR methods to monitor ground deformation, high-resolution InSAR/SAR to study stability of infrastructures, and polarimetric InSAR.

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Tim Wright is a Professor of Satellite Geodesy at the University of Leeds, Leeds, U.K., and the Director of the NERC Centre for the Observation and Modelling of Earthquakes, Volcanoes and Tectonics (COMET). His work has been at the forefront of developing the use of satellite radar interferometry (InSAR) for measuring tectonic and volcanic deformation. Major achievements include the first demonstration that interseismic strain can be measured using InSAR, the investigation of a series of major earthquakes using geodesy, seismology, and geomorphology, and the study of a major rifting episode in Afar, Ethiopia. He has published more than 100 articles in major international journals.

Dr. Wright received the AGU Geodesy Section Award in 2014, the British Geophysical Association's Bullerwell Lecturer in 2015, and the Rosenstiel Award from the University of Miami. He was the Royal Astronomical Society's 2017 Harold Jeffreys Lecturer. In 2018, he co-founded SatSense Ltd.

James Thompson, photograph and biography not available at time of publication.



Alin Marian Achim (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from the "Politehnica" University of Bucharest, Bucharest, Romania, in 1995 and 1996, respectively, and the Ph.D. degree in biomedical engineering from the University of Patras, Patras, Greece, in 2003.

He then obtained the European Research Consortium for Informatics and Mathematics (ERCIM) Post-Doctoral Fellowship, which he spent with the Institute of Information Science and Technologies (ISTI-CNR), Pisa, Italy, and with the French National Institute for Research in Computer Science and Control (INRIA), Sophia Antipolis, France. In October 2004, he joined the Department of Electrical and Electronic Engineering, University of Bristol, Bristol, U.K., as a Lecturer, he became a Senior Lecturer (Associate Professor) in 2010, and a Reader in Biomedical Image Computing in 2015. From August 2018, he holds the Chair in Computational Imaging at the University of Bristol and from 2019 to 2020, he was a Leverhulme Trust Research Fellow with the Laboratoire I3S, Université Cote d'Azur, Nice, France. He has coauthored over 140 scientific publications, including 46 journal articles. His research interest includes statistical signal, image, and video processing with a particular emphasis on the use of sparse distributions within sparse domains and with applications in both biomedical imaging and remote sensing.

Dr. Achim is a Senior Area Editor of the IEEE TRANSACTIONS ON IMAGE PROCESSING, an Associate Editor of the IEEE TRANSACTIONS ON COMPUTATIONAL IMAGING, and an Editorial Board Member of MDPI's *Remote Sensing*. He was/is an elected member of the Bio Imaging and Signal Processing Technical Committee of the IEEE Signal Processing Society, an affiliated member (invited) of the same Society's Signal Processing Theory and Methods Technical Committee and a member of the IEEE Geoscience and Remote Sensing Society's Image Analysis and Data Fusion Technical Committee.



David Bull (Fellow, IEEE) received the B.Sc. degree from the University of Exeter, Exeter, U.K., in 1980, the M.Sc. degree from the University of Manchester, Manchester, U.K., in 1983, and the Ph.D. degree from the University of Cardiff, Cardiff, U.K., in 1988.

He has previously been a Systems Engineer with Rolls Royce, Bristol, U.K., and a Lecturer at the University of Wales, Cardiff. He joined the University of Bristol, Bristol, in 1993, and is its Chair of Signal Processing and the Director of its Bristol Vision Institute. In 2001, he co-founded a university spin-off company, ProVision Communication Technologies Ltd., specializing in wireless video technology. He has authored over 450 articles on the topics of image and video communications and analysis for wireless, Internet, and broadcast applications, together with numerous patents, several of which have been exploited commercially.

Dr. Bull is a fellow of the Institution of Engineering and Technology. He received two IEE Premium Awards. He has also received two IEEE Premium awards for his work. He is the author of three books, and has delivered numerous invited/keynote lectures and tutorials.

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